PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Umess

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Pardume Sharma

2210990638

Supervised By:

Dr. Rajat Takkar

Supervisor

Chitkara university

CHITKARA UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Umess"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Rajat Takkar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Pardume Sharma**

Date:29-04-2026 University Roll No 2210990638

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Rajat Takkar
Professor
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology,
Chitkara University, Punjab, India

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Pardume Sharma
2210990638

Table Of Content

Abstract

The **Mess Management System (Umess)** is a modern web-based application developed to digitize and simplify the management of university mess operations. In many educational institutions, traditional mess management systems still depend on physical mess cards and manual record-keeping processes, which often lead to problems such as card duplication, unauthorized usage, data inaccuracy, and inefficient management. Students also face difficulties in accessing updated mess information, submitting genuine feedback, and communicating effectively with mess authorities. To overcome these challenges, Umess provides a secure, transparent, and technology-driven platform that improves both student experience and administrative efficiency.

The system is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js, ensuring a scalable and responsive application architecture. Additional cloud services such as Cloudinary are used for secure storage of QR codes and media files, while Vercel and Render are utilized for efficient deployment of frontend and backend services.

One of the major features of Umess is its QR code-based authentication system. Each student is assigned a unique QR code linked to their profile/roll number, which can be scanned at the mess counter to verify identity and meal eligibility. This mechanism eliminates the dependency on physical mess cards and significantly reduces the risk of impersonation or misuse. Also it helps to keep a track on the food wastage.

Apart from authentication, the platform provides several student-oriented services. Students can view daily or weekly mess menus, receive instant notifications regarding menu changes or important announcements, and submit anonymous feedback about food quality, hygiene, and service.

For administrators and mess staff, Umess offers role-based access control and dedicated dashboards that simplify operational tasks. Different users, including directors, wardens, guards, and mess staff, can access features relevant to their responsibilities. The system also supports analytical reports and graphical insights related to meal consumption and food wastage, enabling better decision-making and resource management.

Overall, this Umess serves as a comprehensive digital solution that modernizes conventional mess operations through automation, cloud integration, and secure authentication techniques.

Introduction

The **Mess Management System (Umess)** is an innovative digital platform designed to transform the conventional, paper-based processes prevalent in university mess operations into a streamlined, secure, and smart solution. In many universities across India, mess facilities continue to rely on physical mess cards—often made of paper or plastic—which not only pose challenges in terms of maintenance and authenticity but are also prone to misuse and manipulation. Students frequently face issues like lost cards, unauthorized usage by others, and the lack of a proper mechanism to register genuine feedback or access important real-time updates from the mess administration.

This project leverages **modern web technologies and cloud infrastructure**, utilizing the powerful **MERN stack** (MongoDB, Express.js, React.js, Node.js) for robust development, alongside **Cloudinary** for secure QR code and media storage, **Vercel** for front-end deployment, and **Render** for scalable backend services. This combination ensures high performance, secure data handling, and scalable architecture suitable for handling the diverse needs of educational institutions.

At its core, the system is not just a software solution—it is a **digitally-driven ecosystem** tailored to meet the operational, logistical, and feedback-related demands of student dining systems. The system introduces **QR code-based authentication**, generated uniquely for each student and stored on the cloud, eliminating the risk of impersonation or card sharing. Students can simply scan their unique QR code at the mess counter to avail their meals, ensuring that only authorized users are served.

Additionally, it empowers students by providing them with a user-friendly portal to access **real-time mess menus**, receive **instant notifications** about changes or alerts, and submit **anonymous feedback** directly visible to higher-level authorities such as the Director or Warden. This feature ensures accountability and encourages mess staff to maintain higher standards in terms of hygiene, service, and food quality.

From the administrative perspective, the system supports **role-based access control** for Directors, Wardens, Guards, and Mess Staff, ensuring that every stakeholder has access to tailored dashboards and tools. Admins can track **meal consumption trends**, analyze **food wastage through interactive graphs**, and download **daily Excel reports** of food consumption for accountability and analysis.

The system was created with a clear and prepared plan to make certain it works properly and runs smoothly. First, the issues with the old system was the use of mess cards. This blanketed searching at the vintage ways of doing matters, like using paper mess cards, maintaining information on paper, and having sluggish communique strategies. After getting the reviews and scope of this idea, the project was built using the MERN stack with cloud based storage and deployment.

MongoDB was used to store statistics about college students, their meals, remarks, notices, and menus. express.js and Node.js to create the backend logics, inclusive of APIs and server operations. React.js turned into a user-pleasant and responsive website for customers to have interaction with, while a student logs in with their username and password, a unique QR code is created for them and saved adequately on Cloudinary.

When they want to consume, they scan their QR code at the mess counter. The device checks who they are and then statistics that they have eaten or not.

College admins can manage the menu, send notices, and view feedback through unique dashboards designed for distinctive roles.

College students get real-time updates right away. Any anonymous remarks they give, can be reviewed by the higher authorities to help in enhancement of meals and providers. The system also tracks how a good deal of food is being used and shows a view showing how a good deal is being wasted, in conjunction with charts and graphs to display the facts really.

The entire website was launched using Vercel for the front component and Render for the back component, making it clean to develop and get admission to from the cloud.

The Mess manipulate device changed into developed the use of present day net technology and cloud platforms to ensure excessive overall performance, protection, and scalability.

1. MERN Stack

- MongoDB: Used as the NoSQL database to save pupil information, meal records, comments, notices, and menu info.
- Expresse.js: Used for developing backend APIs and
- React.js: Used for creating a responsive and interactive person interface.
- Node.js: Used due to the fact the runtime environment for executing backend services.

2. Cloudinary

Cloudinary is used for storing and dealing with QR codes and media files such as profile pictures, securely in the cloud. It lets in easy retrieval and green cloud-based totally storage management.

3. Vercel

Vercel is used for deploying the frontend React software program. It presents fast performance, continuous deployment, and scalability.

4. Render

Render is used for backend deployment and web hosting server-thing APIs. It ensures dependable backend usual overall performance and cloud scalability.

5. QR Code generation

Specific QR codes are generated for every scholar to authenticate meals availment securely and save you misuse of mess playing cards.

6. Extra functions

- Real-Time Notifications
- Anonymous feedback machine
- Meal Wastage Analytics
- Dashboard and record era.

The **Mess Management System** extends across several domains, providing benefits for students, administrators, and mess staff alike. The project scope includes:

1. User Authentication and Security

- **Login System:** Students log in using their **unique roll number and password**, ensuring a secure and personalized experience.

- **QR Code Generation:** Each student's login credentials are linked to a **unique QR code**, stored on **Cloudinary** for easy retrieval.

2. QR Code-Based Food Availment

- Students scan their **QR code** to avail food. This system prevents any misuse of mess cards and ensures accurate tracking of food consumption.

- The **system records** which student has availed which meal, keeping track of meal types and the number of meals served.

3. Real-Time Menu Updates and Notices

- **Menu Updates:** Mess staff can instantly update the menu and notify students of any changes, whether due to availability or special items.

- **Notice Board:** Administrators can post critical messages regarding mess timings, health updates, or upcoming events, which are automatically visible to all students.

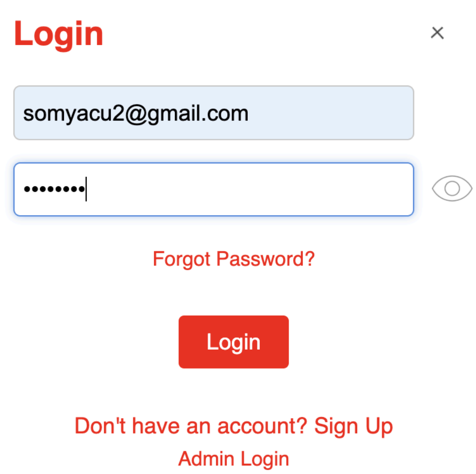
4. Feedback Collection

- **Anonymous Feedback:** Students can submit feedback without revealing their identity, ensuring honest reviews.

- **Review Management:** Admins can view, analyze, and take action based on the feedback provided by students.

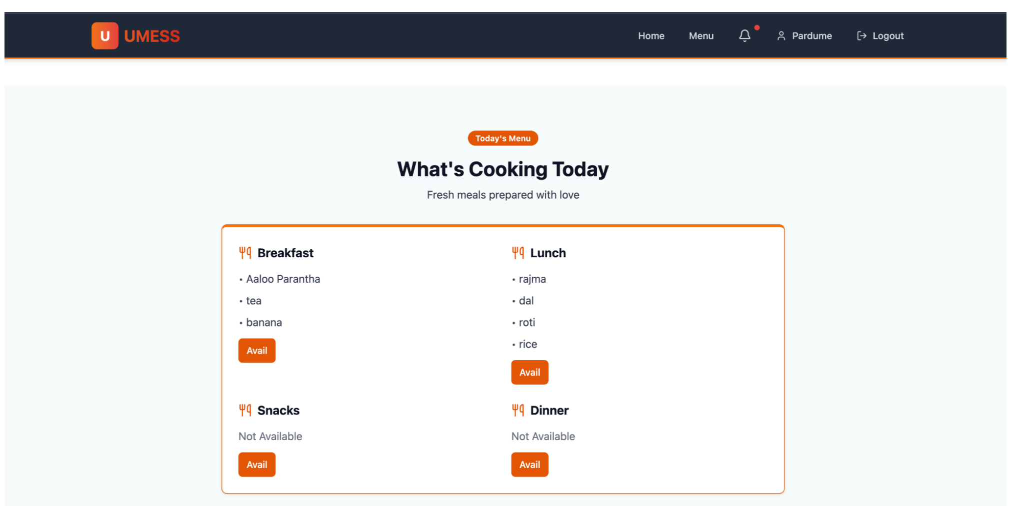
1. Student

Login and Authentication



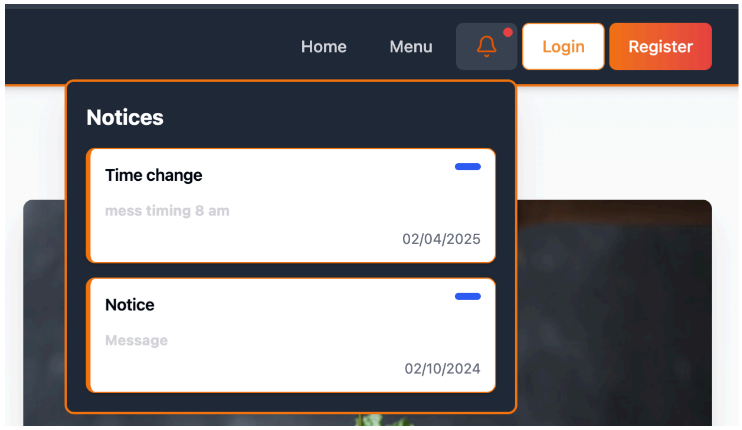
- Students log in with their **unique roll number** and **password**.
- The system generates a **QR code** linked to the student's ID, stored on **Cloudinary**.

Food Availment



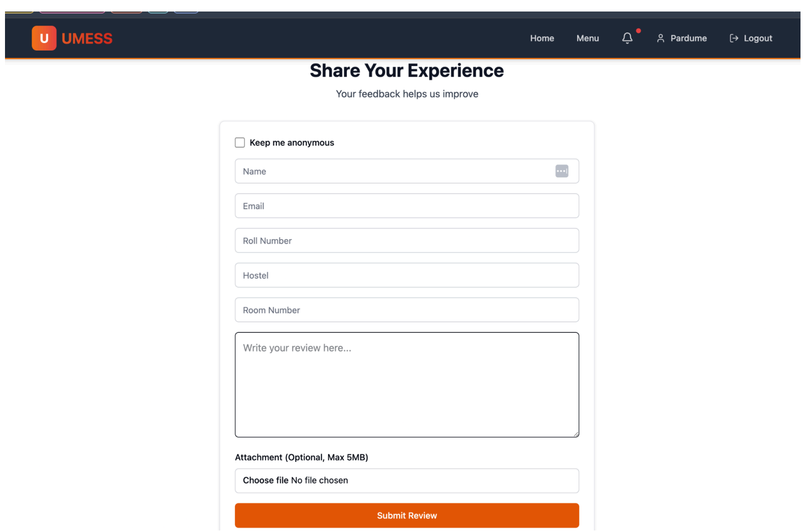
- **QR Code Scan:** At the mess counter, students scan their QR code to avail food.
- The system logs the meal and prevents any misuse by another student.

Real-Time Menu and Notices



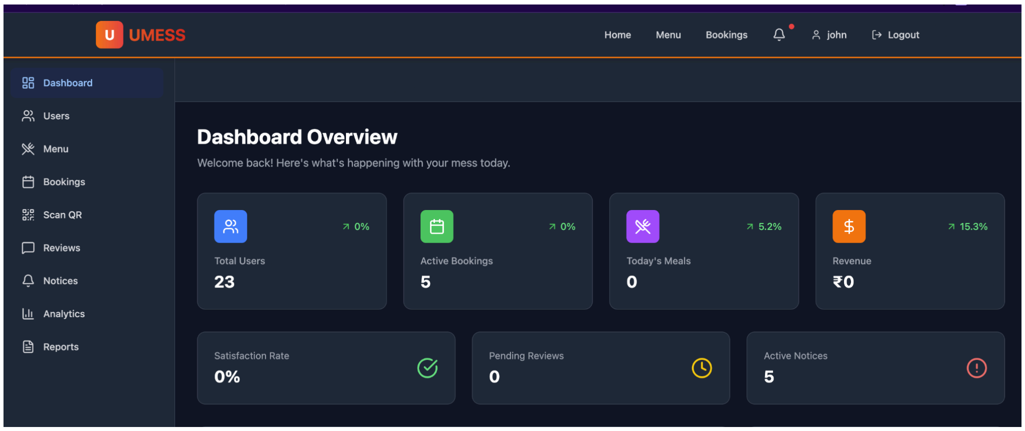
- Students have **access to the updated menu** for each meal (breakfast, lunch, dinner).
- **Real-time notices** are displayed about important updates, such as changes in the schedule or new meal items.

Anonymous Feedback



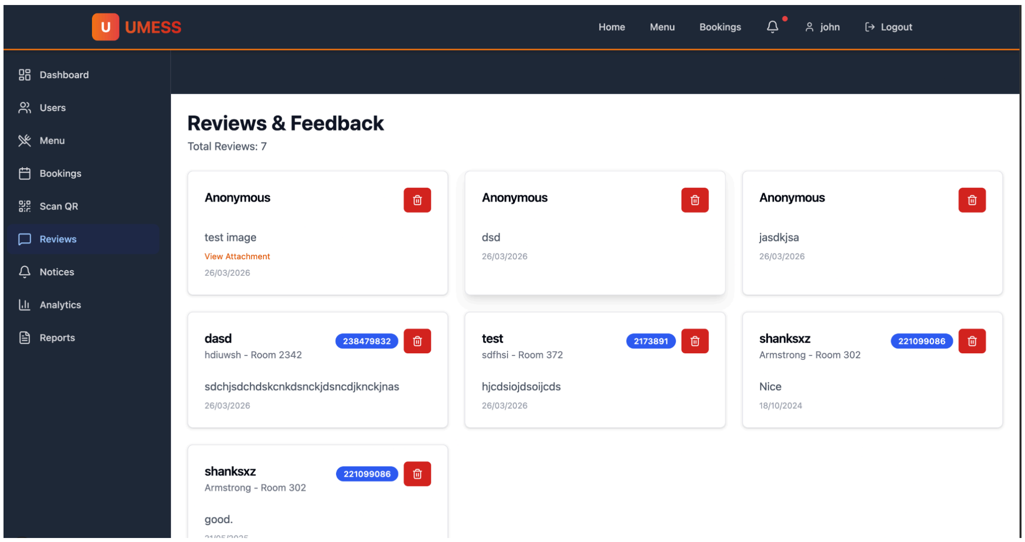
2. Admin

Role-Based Access

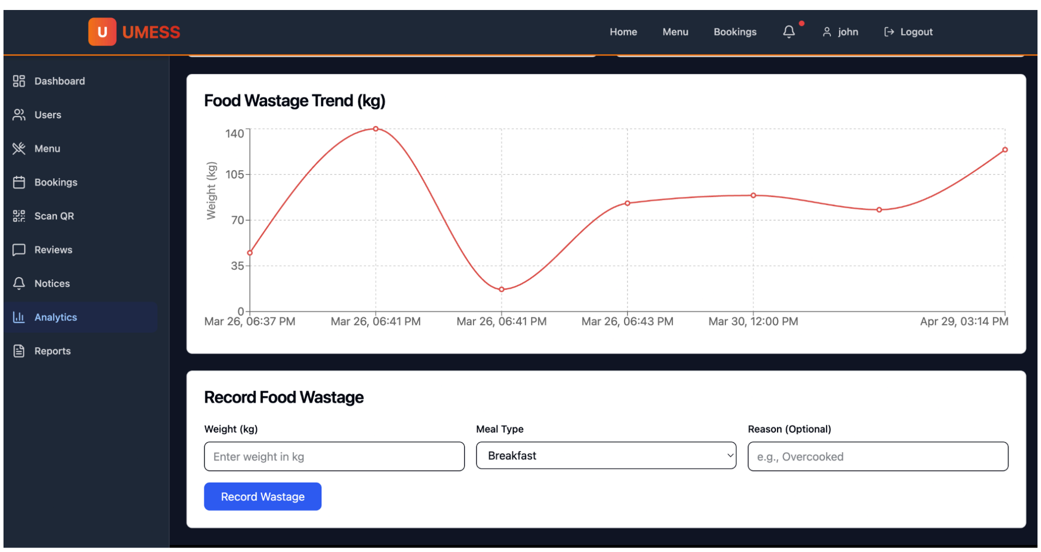


- Different roles (Director, Warden, Staff) have specific permissions.

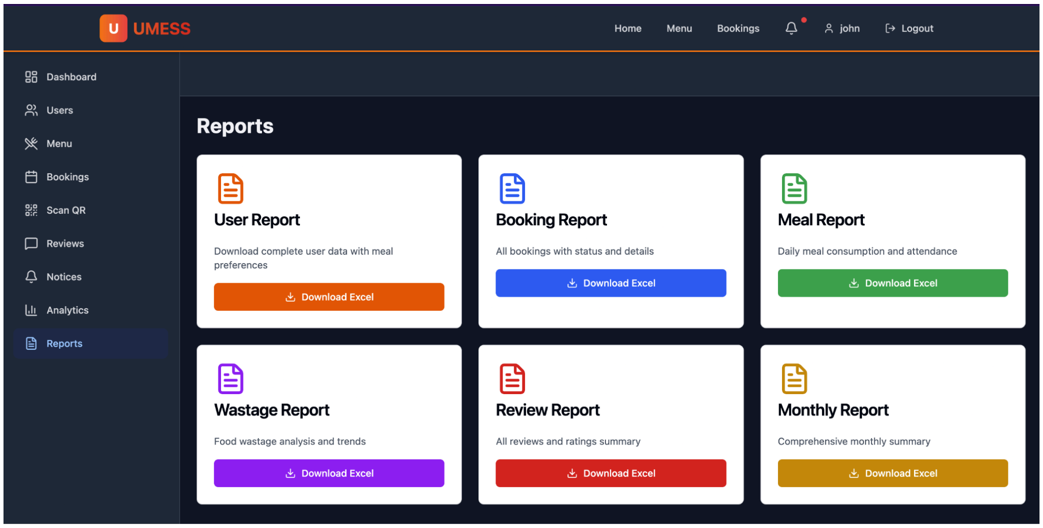
- **Admin Dashboard:** Allows admins to monitor food availability, update menus, and manage notices.
- Reviews can be viewed by the admins.



Food Wastage Analytics

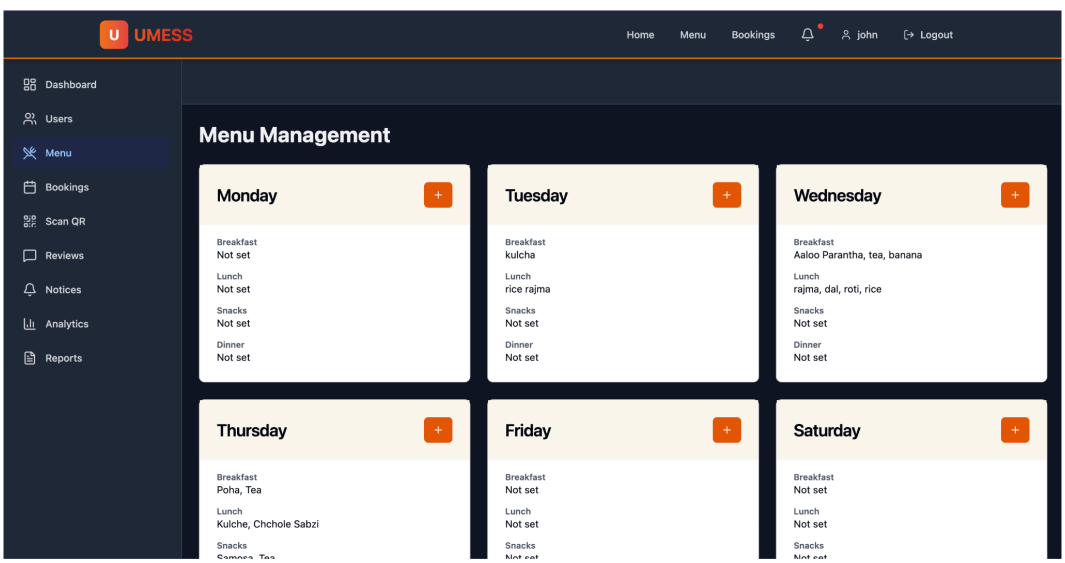


- **Reports and Graphs:** Admins receive data-driven reports on food wastage, helping them optimize meal planning and reduce waste.

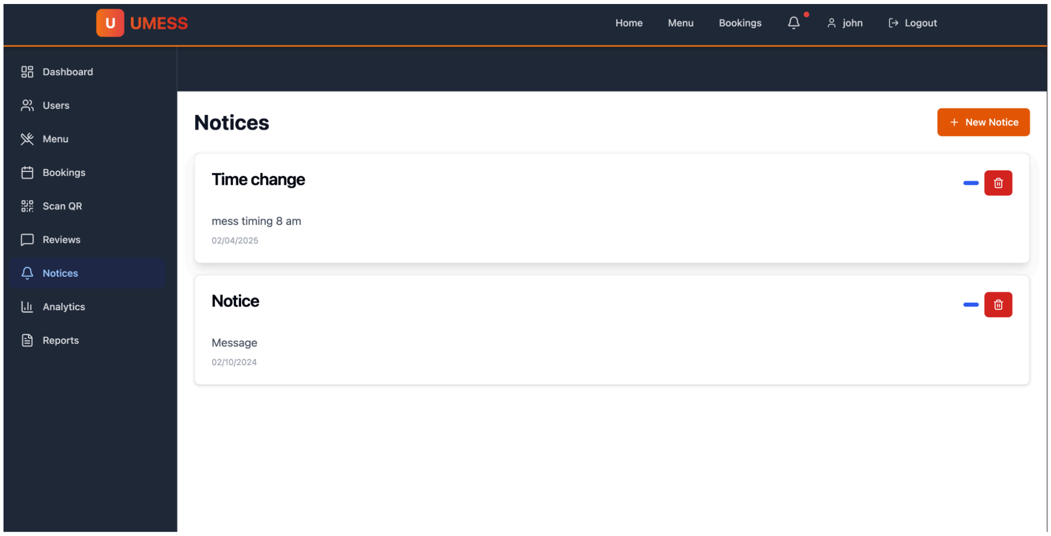


CRUD Operations for Menus and Notices

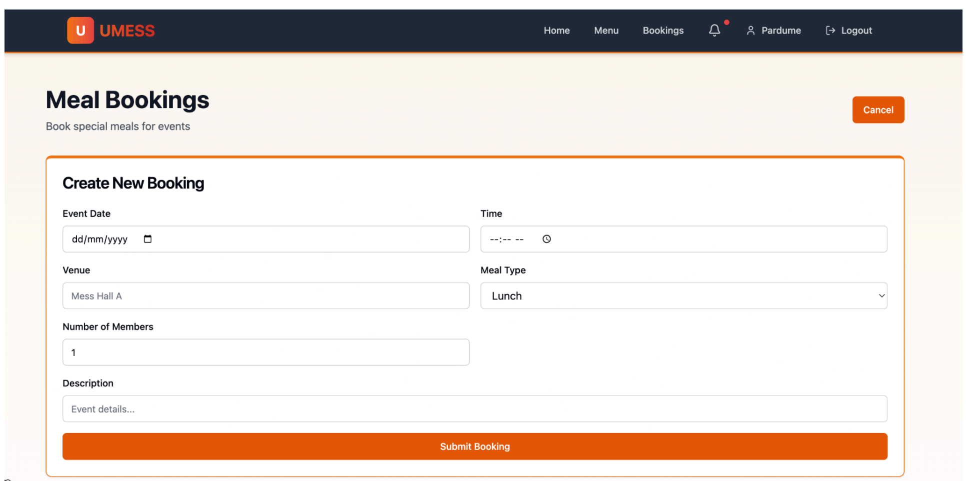
- **CRUD for Menus:** Admins can create, read, update, and delete menu items in real-time.



- **CRUD for Notices:** Critical notices can be easily posted and updated to ensure students are informed immediately.



- **Faculties** can also do the **booking** of the **venues** as per their choice.



The implementation of the Mess management system (Umess) added extremely good upgrades to the general functioning of university mess operations. The system efficiently transformed the conventional guide technique into a cozy, apparent, and digitally controlled environment. via integrating QR code authentication, cloud-primarily based absolute storage, real-time updates, and role-primarily based total management, the challenge finished higher operational performance and stepped forward customer pleasure for every university college students and administrators.

Major Findings

1. Good deal in Misuse of bodily Mess playing gambling cards

One of the most vital findings of the undertaking changed into a successful elimination of troubles related to standard paper-primarily based mess playing cards. University students often experience problems which embody card loss, duplication, and unauthorized utilization through others.

2. Advanced Accuracy in Meal intake tracking

The system enabled computerized digital recording of every meal transaction through QR code scanning. This real-time tracking mechanism advanced the accuracy of meal consumption statistics and removed the want for manual attendance registers or paper logs. Administrators should without issues show the quantity of college students availing food for breakfast, lunch, and dinner, principal to higher operational manipulation and records reliability.

3. Greater appropriate Transparency via anonymous comments

Traditional feedback techniques regularly discouraged college students from expressing right evaluations due to fear of identification or terrible results. The anonymous comments characteristic endorsed students to offer honest reviews concerning food, hygiene, cleanliness, and other necessities. As comments have emerged as a proper way on hand to better authorities collectively with directors and Wardens, it advanced responsibility amongst the mess frame of workers and promoted transparency inside the device.

4. Effective meals Wastage monitoring and Optimization

Some particularly important final consequences of the tool changed into the functionality to reveal meals consumption styles and choose out wastage tendencies. Through analyzing meal availment statistics accumulated through QR scans, directors received insights into the amount of food eaten up each day. These facts helped management to optimize the meal training and reduce useless meals wastage. The analytical critiques and graphical dashboards moreover supported higher planning and aid utilization.

5. progressed communique among college university college students and management

The implementation of actual-time menu updates and be aware manipulate stepped forward communication substantially. Students want to right away get right of entry to up to date menus, time desk modifications, specific bulletins, and important notices via the portal. This decreased confusion related to meal timings or menu modifications and ensured that students remained knowledgeable usually.

6. Extended Administrative performance through the Dashboards

The characteristic-based completely gets admission to manipulate system simplified administrative operations with the useful resource of assigning precise permissions to as a minimum one-of-a-type user which encompass directors, Wardens, Guards, and Mess frame of employees. Each stakeholder also needs to get proper entry to features relevant to their responsibilities.

Outcomes

1. Decreased Dependency on manual and Paper-based without a doubt strategies

The system efficiently changed conventional paper-based mess cards and guide report-maintaining techniques with a digital and automated system. This decreased paperwork, minimized human mistakes, and simplified every day mess operations.

2. Superior safety and Authentication

The QR code authentication mechanism extra protection with the useful resource of ensuring that meals had been availed to the hosteller.

3. Quicker and in addition dependable communication

The real-time notification and menu replacement feature ensured that students acquired accurate statistics properly. Essential announcements regarding menu changes, unique sports, or health recommendations can be communicated proper now.

4. Higher monitoring and statistics evaluation

Administrators get right of access to one-of-a-type reports, graphs, and analytical insights regarding meal consumption and food wastage. This statistics-pushed approach supported better choice-making and operational planning.

5. Scalability and destiny Adaptability

Because the system has emerged as advanced technology and cloud-based services and scalable structure, it may without troubles be prolonged to useful resource big institutions, extra features, and cell systems inside the destiny.

Conclusion

Umess presents a whole virtual answer for modernizing traditional university mess operations. by using the use of changing bodily mess cards with QR code-primarily based authentication, the system improves safety, transparency, and operational performance. capabilities along with real-time menu updates, anonymous remarks series, meals wastage analytics.

The project demonstrates how modern web technologies and cloud infrastructure can effectively be used to resolve the actual administrative problems in educational institutions. This system reduces guide workload, minimizes misuse, improves communication, and helps better decision-making via analytical critiques and dashboards.

Future Scope

- Integration of biometric authentication for extra suitable protection.
- App development for Android and iOS systems.
- AI-primarily based meals demand prediction to in addition lessen food wastage.
- Multi-college assist for big-scale deployment.
- Advanced analytics and reporting the use of system learning strategies.

- **MERN Stack Documentation:** How To Use MERN Stack: A Complete Guide
- **Cloudinary Documentation:** Cloudinary Image & Video Management - Documentation Home
- **Vercel Documentation:** Vercel Documentation
- **Render Documentation:** Docs + Quickstarts - Render
- **QR-Based Canteen Management System Research:** QR-based Canteen Management System - ResearchGate
- **Anonymous Feedback in Education Research:** Using Anonymous Feedback to Increase Student Engagement
- **Food Waste Tracking in Dining Halls Study:** Reducing Food Waste in Campus Dining: A Data-Driven Approach
- **Cloudinary Integration Guide:** Developer Resources for Using Images and Videos in Your Apps
- **Vercel Deployment Guides:** Guides - Vercel
- **Render Deployment Guide:** Render Deployment Guide | The Full-Stack Blog